

Task Performance

Assembly Language Basics

Objective(s):

At the end of the exercise, the students should be able to:

- Determine the role of assembly language in computers; and
- Identify the structure of codes in assembly language.

Instructions:

Part 1 (40 points)

1. Search for two (2) free online assemblers available on the Internet. For each assembler, answer the following items:
 - a. What organization/company manages the online assembler? (2 points)
 - b. Provide the links to the online assembler. (3 points)
 - c. Take a screenshot of the initial content of the code editor of the assembler. (5 points)
 - d. Run the initial codes in the editor and take a screenshot of the output/result. (5 points)
 - e. Does the assembler support an x86 architecture? Elaborate on your answer. (5 points)

Part 2 (30 points)

1. Go to <https://www.jdoodle.com>. At the right portion of the webpage you will see the list of different programming languages (those that are underlined). Click on the **Assembler (NASM)**. This will direct you to the *Online Assembler – NASM Compiler IDE*.
 - The **Netwide Assembler (NASM)** is an assembler that also caters to x86 architecture, such as the Emu8086. It supports a wide range of object file formats including ELF, COFF, Linux and NetBSD/FreeBSD, Microsoft 16-bit OBJ and Win32. It encompasses a simple and easy-to-understand syntax, similar to Intel 8086 but with reduced complexity. NASM was mainly designed for portability and modularity.
2. Click the **Full Screen** button (in line with the **Execute** button) to maximize the assembler's interface. Then, analyze the assembly codes in the code editor. Click the **Execute** button and see the output at the *Result* section.
3. Answer the following question regarding the initial assembly codes in the editor (5 items x 3 points):
 - a. What does the program do?
 - b. Is the result of the program correct?
 - c. How much memory was utilized in the program execution?
 - d. How long did it take to execute the program?
 - e. Under what section were the x and y initialized?
4. Delete the content of the code editor and type the assembly codes below:

```
1 section .text
2     global _start
3
4 _start:
5     mov ax, '9'
6     sub ax, '0'
7
8     mov bl, '3'
9     sub bl, '0'
10
11    div bl
12    add ax, '0'
```

```

13
14     mov [res], ax
15     mov ecx, message
16     mov edx, length
17     mov ebx, 1      ;file descriptor (stdout)
18     mov eax, 4      ;system call number (sys_write)
19     int 0x80        ;call kernel
20
21     mov ecx, res
22     mov edx, 1
23     mov ebx, 1      ;file descriptor (stdout)
24     mov eax, 4      ;system call number (sys_write)
25     int 0x80        ;call kernel
26     mov eax, 1      ;system call number (sys_exit)
27     int 0x80        ;call kernel
28
29 ▾ section .data
30     message db "The result is: "
31     length equ $- message
32
33 ▾ section .bss
34     res resb 2

```

5. Execute the program. Then, answer the following items (5 items x 3 points):
- What does the program do?
 - How many sections were used in the program?
 - Identify at least three (3) mnemonics (opcode/directive/macro) that were used in the program.
 - Identify at least three (3) operands that were used in the program.
 - In the program above, which lines of codes indicate the storage space allocation for the message and message length?

Grading Rubric for Essay:

Criteria	Performance Indicator	Points
Content	Correct ideas, concepts, and/or examples were included.	3
Organization	The ideas were presented in an organized manner.	2
TOTAL		5

Note: DO NOT COPY AND PASTE. Students who copy and paste their work from any website or their classmate will automatically receive a failing mark for this assessment task.